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PHY 421 — Past Questions & Materials
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Paper 1 of 4 — PHY421 ASSIGNMENT-1

Level 400 • Second Semester • PastQuestion

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Course Code: PHY 421

Assignment

Do a term paper on Stern - Gerlach experiment

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ABSTRACT

In 1922, Otto Stern and Walther Gerlach performed a groundbreaking experiment that revolutionized

our understanding of atomic structure and quantum mechanics. By passing a beam of silver atoms through an inhomogeneous magnetic field, they demonstrated the quantization of spin, magnetic moment, and angular momentum, providing conclusive evidence for the wave-particle duality of matter.

This experiment not only confirmed the predictions of quantum theory but also laid the foundation for

significant advancements in fields such as magnetic resonance, spintronics, and quantum computing.

This paper provides an in-depth examination of the Stern-Gerlach experiment, its theoretical framework,

experimental design, results, and far-reaching implications, highlighting its enduring impact on our understanding of the quantum world.

INTRODUCTION

In 1922, Otto Stern and Walther Gerlach performed a landmark experiment that demonstrated the principles of quantum mechanics and revolutionized our understanding of atomic structure and behavior (Stern & Gerlach, 1922). The Stern-Gerlach experiment showed that atoms have an intrinsic

angular momentum, known as spin, which is quantized and can only take on specific discrete values

(Feynman, 1965). This experiment not only confirmed the predictions of quantum theory but also laid

the foundation for significant advancements in fields such as magnetic resonance, spintronics, and quantum computing.

Background

In the early 1920s, the atomic model proposed by Niels Bohr was widely accepted, but it failed to explain the behavior of atoms in magnetic fields (Bohr, 1913). The Stern-Gerlach experiment was designed to test the hypothesis that atoms have a magnetic moment, which is a measure of their tendency to align with a magnetic field (Gerlach, 1921).

The Experiment

The Stern-Gerlach experiment consisted of passing a beam of silver atoms through an inhomogeneous magnetic field and detecting the deflection of the atoms on a screen (Stern & Gerlach, 1922).

RESULTS

The results of the experiment showed that the silver atoms were deflected into two distinct beams, indicating that they had a magnetic moment and were aligning with the magnetic field (Stern & Gerlach, 1922). This was unexpected, as classical physics predicted that the atoms would be deflected continuously, not in discrete beams.

METHODOLOGY

- Vaporize silver atoms using an electric furnace in a vacuum.
- Direct the resulting beam of silver atoms through an inhomogeneous magnetic field.

IMPLICATIONS

The Stern-Gerlach experiment:

1.

Quantization Evidence:

2.

1. It provides direct evidence of quantization in quantum mechanics.
2. The discrete deflection points on the detector screen demonstrate that certain properties (like spin) can only take specific quantized values.

3.

Foundation for Electron Spin:

4.

1. The experiment laid the groundwork for understanding electron spin.
2. Stern and Gerlach's work eventually led to the realization that electrons possess intrinsic angular momentum (spin).

1.

Quantum Measurement Demonstration:

2.

1. It stands as one of the best demonstrations of quantum measurement.
2. The experiment shows how measurement outcomes are inherently probabilistic in quantum systems.

DISCUSSION

The Stern-Gerlach experiment's findings challenged classical understanding, sparking debate and driving innovation in quantum theory. Its impact extends to modern applications in physics, chemistry, and engineering.

CONCLUSION

The Stern-Gerlach experiment marked a significant turning point in physics, providing conclusive evidence for quantum mechanics and shaping our understanding of atomic and subatomic behavior. Its legacy continues to influence scientific inquiry and discovery.

REFERENCES

- Bohr, N. (1913). On the constitution of atoms and molecules. *Philosophical Magazine*, 26(153), 1-25.
- Dirac, P. A. M. (1928). The quantum theory of the electron. *Proceedings of the Cambridge Philosophical Society*, 24, 136-144.
- Feynman, R. P. (1965). *The Feynman lectures on physics*. Addison-Wesley.
- Gerlach, W. (1921). Über die Richtungsquantelung im Magnetfeld. *Zeitschrift für Physik*, 5(2), 189-204.
- Pauli, W. (1940). The spin-statistics theorem. *Physical Review*, 58(8), 716-722.
- Stern, O., & Gerlach, W. (1922). Der experimentelle Nachweis der Richtungsquantelung im Magnetfeld. *Zeitschrift für Physik*, 9(4), 349-355.

Paper 2 of 4 — PHY421 ASSIGNMENT

Level 400 • Second Semester • PastQuestion

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Course Code: PHY 421

Assignment

Do a term paper on Stern - Gerlach experiment

TABLE OF CONTENTS

- Abstract
- Introduction
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ABSTRACT

The Stern-Gerlach experiment, conducted in 1922, was a groundbreaking demonstration of quantum mechanics.

By measuring the deflection of silver atoms in a non-uniform magnetic field, Otto Stern and Walther Gerlach

revealed the quantization of angular momentum and wave-particle duality. This paper provides an in-depth

analysis of the experiment's background, methodology, results, and implications, highlighting its significance in the

development of quantum theory.

INTRODUCTION

In the early 20th century, physicists struggled to understand the nature of matter and energy. The Stern-Gerlach

experiment addressed this challenge, providing conclusive evidence for quantum mechanics and shaping our

understanding of atomic and subatomic behavior.

BACKGROUND

Niels Bohr's atomic model (1913) proposed energy quantization, but wave-particle duality remained unclear. The

Stern-Gerlach experiment aimed to measure the magnetic moment of silver atoms, expected to be continuous.

METHODOLOGY

Stern and Gerlach designed an apparatus consisting of:

1. An oven to produce a beam of silver atoms
2. A non-uniform magnetic field to deflect the atoms
3. A detector to measure deflection patterns

RESULTS

The experiment revealed two distinct beams of silver atoms, deflected at specific angles, indicating quantized

angular momentum. This contradicted classical expectations and supported Bohr's atomic model.

IMPLICATIONS

The Stern-Gerlach experiment:

1. Demonstrated wave-particle duality, showing particles can exhibit both wave-like and particle-like behavior.
2. Confirmed the quantization of angular momentum, supporting Bohr's atomic model.
3. Paved the way for quantum mechanics, influencing Schrödinger's equation and Heisenberg's uncertainty principle.

DISCUSSION

The Stern-Gerlach experiment's findings challenged classical understanding, sparking debate and driving

innovation in quantum theory. Its impact extends to modern applications in physics, chemistry, and engineering.

CONCLUSION

The Stern-Gerlach experiment marked a significant turning point in physics, providing conclusive evidence for

quantum mechanics and shaping our understanding of atomic and subatomic behavior. Its legacy continues to

influence scientific inquiry and discovery.

REFERENCES

Stern, O., & Gerlach, W. (1922). Das magnetische Moment des Silberatoms. Zeitschrift für Physik, 8(3), 110-111.

Bohr, N. (1913). On the Constitution of Atoms and Molecules. Philosophical Magazine, 26(153), 1-25.

Heisenberg, W. (1927). Über den anschaulichen Inhalt der quantentheoretischen Kinematik und Mechanik.

Zeitschrift für Physik, 43(3-4), 167-181.

Paper 3 of 4 — PHY 421 PQ & ANS (1) QUANTUM MECHANICS II

Level 400 • Second Semester • PastQuestion

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Federal University of Petroleum Resources Effurun, Nigeria.
College of Science
Department of Physics
Second Semester Examination

Academic Session: 2019/2020
Course Title: Quantum Mechanics II
Course Code: PHY421 (3 units)
Time Allowed: 3hrs.

Date: 13/07/2021

Instructions: Answer Question 1 and Any Other Three Questions. Complete and ordered answers are preferred to fragments. Note that all symbols have their usual meanings.

Question 1

- a) Consider two particles of masses m_1 and m_2 , interacting via a time-independent potential $V(r_1 - r_2)$ which depends only upon the relative coordinate $r_1 - r_2$. The classical Hamiltonian is therefore

$$H = \frac{p_1^2}{2m_1} + \frac{p_2^2}{2m_2} + V(r_1 - r_2).$$

By using $P_1 = -i\hbar\nabla_1$, $P_2 = -i\hbar\nabla_2$, show that the time-dependent Schrodinger equation with the reduced mass μ of the two particles is given by

$$i\hbar \frac{\partial \psi(r, R, r)}{\partial t} = \frac{\hbar^2}{2\mu} \frac{\partial^2 \psi(r, R, r)}{\partial r^2} - \frac{\hbar^2}{M} \frac{\partial^2 \psi(r, R, r)}{\partial R^2} + V(r) \psi(r, R, r).$$

Where $M = m_1 + m_2$ and $\mu = (m_1 m_2 / (m_1 + m_2))$.

- b) Also, using the relative momentum $p = \frac{m_2 p_1 - m_1 p_2}{m_1 + m_2}$, together with the total momentum

$P = p_1 + p_2$, show that the quantum mechanical Hamiltonian operator is given by

$$H = -\frac{\hbar^2}{2M} \nabla_R^2 - \frac{\hbar^2}{2\mu} \nabla_r^2 + V(r).$$

Question 2

- a) i. State the quantum conditions and the properties of the commutator.
ii. Use quantum conditions and the properties of the commutator to establish the following commutation relations: $[S_x, S_y] = i\hbar S_z$, $[S_y, S_z] = i\hbar S_x$, $[S_z, S_x] = i\hbar S_y$.

- b) Using the Pauli's matrices show that

$$(\vec{\sigma} \cdot \vec{A})(\vec{\sigma} \cdot \vec{B}) = (\vec{A} \cdot \vec{B})I + i\vec{\sigma} \cdot (\vec{A} \times \vec{B}).$$

Where $\vec{\sigma} = (\sigma_x, \sigma_y, \sigma_z)$, $\vec{A} = (A_x, A_y, A_z)$, $\vec{B} = (B_x, B_y, B_z)$ and I is a 2×2 unit matrix.

$$\text{ii } \exp\left[-\frac{i\theta}{\hbar} \vec{n} \cdot \vec{\sigma}\right] = \cos\left(\frac{\theta}{2}\right)I - i\vec{n} \cdot \vec{\sigma} \sin\left(\frac{\theta}{2}\right)$$

Recall that we can expand an operator $\hat{A}$ in a

$$\text{Taylor Series } e^{-\hat{A}} = \sum \frac{(-1)^n}{n!} \hat{A}^n$$

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✓ Question 3

Given the basis ket $|\alpha_i\rangle$, $i=1, 2, 3$, of the observable A and the ket constitute an orthonormal set.

- If $B = I$ is a unit operator, find the matrix representation of B in the A basis.
- Find the matrix representation of A in its own basis.
- If C is real observable, find the matrix representation of C in the A basis, and show that $\langle\alpha_i|C|\alpha_k\rangle = \langle\alpha_k|C|\alpha_i\rangle$. Prove that if $i=k$, all the diagonal elements of C are real, and each of the other elements is equal to the complex conjugate of its mirror image in the diagonal, the matrix is then called the adjoint matrix.
- If $D = BC$, establish that $\langle\alpha_i|D|\alpha_k\rangle = \sum_j \langle\alpha_i|C|\alpha_j\rangle \langle\alpha_j|B|\alpha_k\rangle$ which is just the law of matrix multiplication. The multiplication of matrices is non-commutative just like the multiplication of linear operators.

✓ Question 4

- What do you understand by identical particles?
- Write down the symmetric and antisymmetric properties for any arbitrary permutation σ of identical particles. Give one example each of particles with completely symmetrical wave functions and completely antisymmetrical wave functions.
- Use the symmetrization postulate for fermions to derive the Pauli exclusion principle.
- Show explicitly that Slater's determinant for two particles (fermions) is antisymmetric.

✓ Question 5

- If $a = a(q, p)$ and $a^\dagger = a^\dagger(q, p)$ be the lowering and raising operators of a one-dimensional simple Harmonic Oscillator. Find q and p in terms of a and $a^\dagger$.
- Find the commutators (i) $[a, a] = 0$ (ii) $[a^\dagger, a^\dagger] = 0$ and (iii) $[a, a^\dagger] = 1$.
- Show that the Hamiltonian of a one-dimensional simple Harmonic oscillator is $H = (N + \frac{1}{2})\omega\hbar$, where $N = a^\dagger a$, and
- Show that $[a, N] = a$, (ii) $[a^\dagger, N] = a^\dagger$.
- For the energy eigenvalue problem of the Harmonic Oscillator, obtain the eigenvalue of the Hamiltonian in the following.
(i) $H(a|E\rangle)$ (ii) $H(a^\dagger|E\rangle)$.

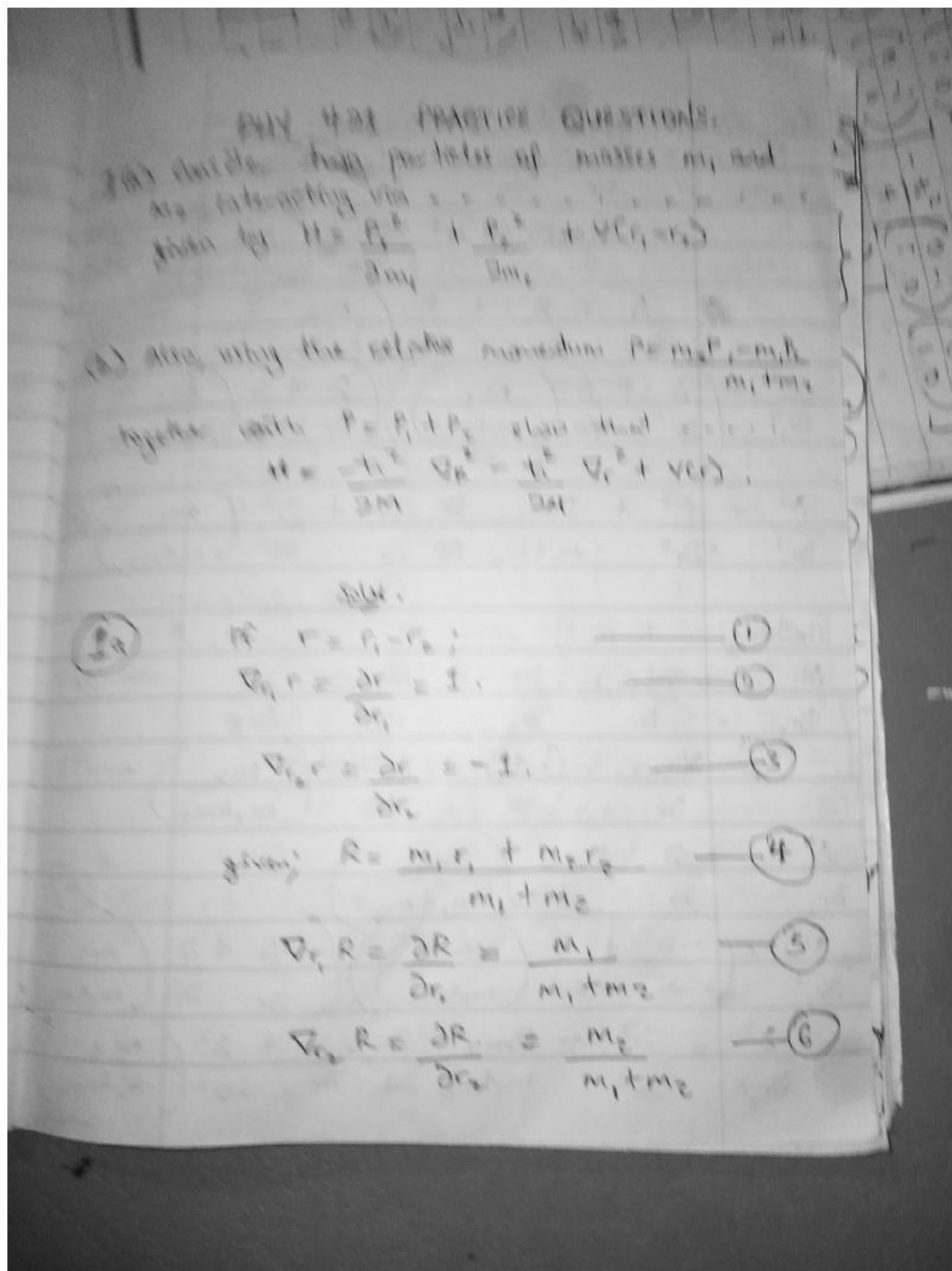
✓ Question 6

- Write down the Hamiltonian for the approximate perturbation theory and define all the terms.
- Derive the formulas for the first- and second-order energy corrections for a time-independent perturbation. Also derive the first-order corrections to the Eigen states. Assume that there is no degeneracy.



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(e, t)



PHY 421 PRACTICE QUESTIONS.

1(a) Consider two particles of masses m_1 and m_2 interacting via - - - - -
 given by $H = \frac{p_1^2}{2m_1} + \frac{p_2^2}{2m_2} + V(r_1 - r_2)$

(b) Also, using the relative momentum $P = \frac{m_2 p_1 - m_1 p_2}{m_1 + m_2}$

together with $P = p_1 + p_2$ show that - - - - -

$$H = \frac{-\hbar^2}{2M} \nabla_R^2 - \frac{\hbar^2}{2\mu} \nabla_r^2 + V(r)$$

Solve.

1a) If $r = r_1 - r_2$; ———— (1)

$\nabla_{r_1} r = \frac{\partial r}{\partial r_1} = 1$. ———— (2)

$\nabla_{r_2} r = \frac{\partial r}{\partial r_2} = -1$. ———— (3)

given; $R = \frac{m_1 r_1 + m_2 r_2}{m_1 + m_2}$ ———— (4)

$\nabla_{r_1} R = \frac{\partial R}{\partial r_1} = \frac{m_1}{m_1 + m_2}$ ———— (5)

$\nabla_{r_2} R = \frac{\partial R}{\partial r_2} = \frac{m_2}{m_1 + m_2}$ ———— (6)

for any arbitrary function; $f(r, r_2)$;

$$\frac{\partial f(r, r_2)}{\partial r_1} = \frac{\partial f(r, R)}{\partial r} \cdot \frac{\partial r}{\partial r_1} + \frac{\partial f(r, R)}{\partial R} \cdot \frac{\partial R}{\partial r_1} \quad (7)$$

$$= \frac{\partial f(r, R)}{\partial r} (1) + \frac{\partial f(r, R)}{\partial R} \left(\frac{m_1}{m_1 + m_2} \right) \quad (8)$$

$$\frac{\partial}{\partial r_1} = \frac{\partial}{\partial r} + \frac{\partial}{\partial R} \left(\frac{m_1}{m_1 + m_2} \right) \quad (9)$$

$$\frac{\partial^2}{\partial r_1^2} = \left(\frac{\partial}{\partial r} + \frac{\partial}{\partial R} \left(\frac{m_1}{m_1 + m_2} \right) \right) \left(\frac{\partial}{\partial r} + \frac{\partial}{\partial R} \left(\frac{m_1}{m_1 + m_2} \right) \right) \quad (10)$$

$$\frac{\partial^2}{\partial r_1^2} = \frac{\partial^2}{\partial r^2} + \frac{2m_1}{m_1 + m_2} \frac{\partial}{\partial R} \cdot \frac{\partial}{\partial r} + \frac{\partial^2}{\partial R^2} \left(\frac{m_1}{m_1 + m_2} \right)^2 \quad (11)$$

similarly;

$$\frac{\partial f(r, r_2)}{\partial r_2} = \frac{\partial f(r, R)}{\partial r} \cdot \frac{\partial r}{\partial r_2} + \frac{\partial f(r, R)}{\partial R} \cdot \frac{\partial R}{\partial r_2} \quad (12)$$

$$= \frac{\partial f(r, R)}{\partial r} \cdot (-1) + \frac{\partial f(r, R)}{\partial R} \left(\frac{m_2}{m_1 + m_2} \right) \quad (13)$$

$$\frac{\partial}{\partial r_2} = -\frac{\partial}{\partial r} + \frac{\partial}{\partial R} \left(\frac{m_2}{m_1 + m_2} \right) \quad (14)$$

$$\frac{\partial^2}{\partial r_2^2} = \left(-\frac{\partial}{\partial r} + \frac{\partial}{\partial R} \left(\frac{m_2}{m_1 + m_2} \right) \right) \left(-\frac{\partial}{\partial r} + \frac{\partial}{\partial R} \left(\frac{m_2}{m_1 + m_2} \right) \right) \quad (15)$$

$$\frac{\partial^2}{\partial r_2^2} = \frac{\partial^2}{\partial r^2} - \frac{2m_2}{m_1 + m_2} \frac{\partial}{\partial R} \cdot \frac{\partial}{\partial r} + \frac{\partial^2}{\partial R^2} \left(\frac{m_2}{m_1 + m_2} \right)^2$$

recall that our seven-dimension time dependent schrodinger equation is:

$$i\hbar \frac{\partial \Psi(r_1, r_2, t)}{\partial t} = \left[\frac{-\hbar^2}{2m_1} \nabla_{r_1}^2 - \frac{\hbar^2}{2m_2} \nabla_{r_2}^2 + V(r_1, r_2) \right] \Psi(r_1, r_2, t) \quad (16)$$

Substituting equ (11) & (15) into equ (16) yields;

$$\begin{aligned} i\hbar \frac{\partial \Psi(r_1, r_2, t)}{\partial t} = & \left[\frac{-\hbar^2}{2m_1} \left[\frac{\partial^2}{\partial r^2} + \frac{2m_1}{m_1 + m_2} \frac{\partial}{\partial R} \frac{\partial}{\partial r} + \frac{\partial^2}{\partial R^2} \left(\frac{m_1}{m_1 + m_2} \right)^2 \right] \right. \\ & \left. - \frac{\hbar^2}{2m_2} \left[\frac{\partial^2}{\partial r^2} - \frac{2m_2}{m_1 + m_2} \frac{\partial}{\partial R} \frac{\partial}{\partial r} + \frac{\partial^2}{\partial R^2} \left(\frac{m_2}{m_1 + m_2} \right)^2 \right] \right] \Psi(r_1, r_2, t) + V(r_1, r_2) \quad (17) \end{aligned}$$

Re-arranging equ (17);

$$\begin{aligned} i\hbar \frac{\partial \Psi(r_1, r_2, t)}{\partial t} = & \left[\frac{-\hbar^2}{2m_1} \left[\frac{\partial^2}{\partial r^2} + \left(\frac{m_1}{m_1 + m_2} \right)^2 \frac{\partial^2}{\partial R^2} + \frac{2m_1}{m_1 + m_2} \frac{\partial}{\partial R} \frac{\partial}{\partial r} \right] \right. \\ & \left. - \frac{\hbar^2}{2m_2} \left[\frac{\partial^2}{\partial r^2} + \left(\frac{m_2}{m_1 + m_2} \right)^2 \frac{\partial^2}{\partial R^2} - \frac{2m_2}{m_1 + m_2} \frac{\partial}{\partial R} \frac{\partial}{\partial r} \right] \right] \Psi(r_1, r_2, t) + V(r_1, r_2) \quad (18) \end{aligned}$$

~~$$i\hbar \frac{\partial \Psi}{\partial t} = \frac{-\hbar^2}{2} \left(\frac{1}{m_1} + \frac{1}{m_2} \right) \frac{\partial^2}{\partial R^2} - \frac{\hbar^2}{2} \left(\frac{m_1}{m_1 + m_2} \right)^2 \frac{\partial^2}{\partial R^2} + \frac{\hbar^2}{2} \left(\frac{m_2}{m_1 + m_2} \right)^2 \frac{\partial^2}{\partial R^2} + \frac{2m_1}{m_1 + m_2} \frac{\partial}{\partial R} \frac{\partial}{\partial r} - \frac{2m_2}{m_1 + m_2} \frac{\partial}{\partial R} \frac{\partial}{\partial r} + V(r_1, r_2) \Psi(r_1, r_2, t)$$~~

$$\frac{i\hbar \partial \Psi}{\partial t} = \left[-\frac{\hbar^2}{2} \left(\frac{1}{m_1} + \frac{1}{m_2} \right) \frac{\partial^2}{\partial r^2} - \frac{\hbar^2}{2} \left[\frac{1}{m_1} \left(\frac{m_1}{m_1+m_2} \right)^2 + \frac{1}{m_2} \left(\frac{m_2}{m_1+m_2} \right)^2 \right] \frac{\partial^2}{\partial R^2} \right] \Psi(r,t) \quad (1b)$$

$$\left\{ -\frac{\hbar^2}{2m_1} \left[\frac{2m_1}{m_1+m_2} \right] \frac{\partial^2}{\partial r^2} + \frac{\hbar^2}{2m_2} \left[\frac{2m_2}{m_1+m_2} \right] \frac{\partial^2}{\partial R^2} \right\} \Psi(r,t) \quad (19)$$

$$\frac{i\hbar \partial \Psi}{\partial t} = \left[-\frac{\hbar^2}{2} \left[\frac{1}{m_1} + \frac{1}{m_2} \right] \frac{\partial^2}{\partial r^2} - \frac{\hbar^2}{2} \left[\frac{m_1}{m_1+m_2} + \frac{m_2}{m_1+m_2} \right] \frac{\partial^2}{\partial R^2} + V(r_1-r_2) \right] \Psi(r,t) \quad (20)$$

$$\text{let } (M = m_1 + m_2) \text{ \& } \left(\mu = \frac{m_1 m_2}{m_1 + m_2} \right); \quad (21)$$

$$\frac{i\hbar \partial \Psi}{\partial t} = \left\{ -\frac{\hbar^2}{2} \left[\frac{m_1+m_2}{m_1 m_2} \right] \frac{\partial^2}{\partial r^2} - \frac{\hbar^2}{2} \left[\frac{m_1+m_2}{(m_1+m_2)^2} \right] \frac{\partial^2}{\partial R^2} + V(r_1-r_2) \right\} \Psi(r,t) \quad (21')$$

$$\frac{i\hbar \partial \Psi}{\partial t} = -\frac{\hbar^2}{2} \left(\frac{1}{\mu} \right) \frac{\partial^2}{\partial r^2} - \frac{\hbar^2}{2} \left(\frac{1}{M} \right) \frac{\partial^2}{\partial R^2} + V(r_1-r_2) \quad (22)$$

$$\therefore \frac{i\hbar \partial \Psi(r,R,t)}{\partial t} = \left[-\frac{\hbar^2}{2\mu} \frac{\partial^2}{\partial r^2} - \frac{\hbar^2}{2M} \frac{\partial^2}{\partial R^2} + V(r_1-r_2) \right] \Psi(r,R,t) \quad (23)$$

equ (23) is the time independent schrodinger equation with reduced mass; $\left[\mu = \frac{m_1 m_2}{m_1 + m_2} \text{ and } \frac{1}{M} = \frac{1}{m_1 + m_2} \right]$

using $P = \frac{m_2 P_1 - m_1 P_2}{m_1 + m_2}$

$P = P_1 + P_2$

soln.

If $P = P_1 + P_2$ — (1)

$\nabla_{P_1} P = 1$; $\nabla_{P_2} P = 1$ — (2)

similarly;

$R = \frac{m_2 P_1 - m_1 P_2}{m_1 + m_2}$ — (3)

$\nabla_{P_1} R = \frac{m_2}{m_1 + m_2}$ — (4)

$\nabla_{P_2} R = \frac{-m_1}{m_1 + m_2}$ — (4)

$\frac{\partial f_{C,P,R}}{\partial P_1} = \frac{\partial f}{\partial P} \cdot \frac{\partial P}{\partial P_1} + \frac{\partial f}{\partial R} \cdot \frac{\partial R}{\partial P_1}$ — (5)

$= \frac{\partial f}{\partial P} (1) + \frac{\partial f}{\partial R} \left(\frac{m_2}{m_1 + m_2} \right)$ — (6)

$\frac{\partial}{\partial P_1} = \frac{\partial}{\partial P} + \left(\frac{m_2}{m_1 + m_2} \right) \frac{\partial}{\partial R}$ — (7)

$\frac{\partial^2}{\partial P_1^2} = \left(\frac{\partial}{\partial P} + \left(\frac{m_2}{m_1 + m_2} \right) \frac{\partial}{\partial R} \right) \left(\frac{\partial}{\partial P} + \left(\frac{m_2}{m_1 + m_2} \right) \frac{\partial}{\partial R} \right)$ — (8)

$\frac{\partial^2}{\partial P_1^2} = \frac{\partial^2}{\partial P^2} + \left(\frac{m_2}{m_1 + m_2} \right)^2 \frac{\partial^2}{\partial R^2} + \frac{2m_2}{m_1 + m_2} \frac{\partial}{\partial P} \frac{\partial}{\partial R}$ — (9)

Similarly:

$$\frac{\partial f(R, P, t)}{\partial P_2} = \frac{\partial f}{\partial P} \frac{\partial P}{\partial P_2} + \frac{\partial f}{\partial R} \frac{\partial R}{\partial P_2} \quad (10)$$

$$\frac{\partial f}{\partial P_2} = \frac{\partial f}{\partial P} (1) + \frac{\partial f}{\partial R} \left(\frac{-m_1}{m_1 + m_2} \right) \quad (11)$$

$$\frac{\partial}{\partial P_2} = \frac{\partial}{\partial P} - \frac{m_1}{m_1 + m_2} \frac{\partial}{\partial R} \quad (12)$$

$$\frac{\partial^2}{\partial P_2^2} = \left(\frac{\partial}{\partial P} - \frac{m_1}{m_1 + m_2} \frac{\partial}{\partial R} \right) \left(\frac{\partial}{\partial P} - \frac{m_1}{m_1 + m_2} \frac{\partial}{\partial R} \right)$$

$$\frac{\partial^2}{\partial P_2^2} = \frac{\partial^2}{\partial P^2} + \left(\frac{m_1}{m_1 + m_2} \right)^2 \frac{\partial^2}{\partial R^2} - \frac{2m_1}{m_1 + m_2} \frac{\partial}{\partial R} \frac{\partial}{\partial P} \quad (13)$$

recall our seven-time time dependent schrodinger equation is given as;

$$i\hbar \frac{\partial \Psi(R, P, t)}{\partial t} = \left[\frac{-\hbar^2}{2m_1} \frac{\partial^2}{\partial P_1^2} - \frac{\hbar^2}{2m_2} \frac{\partial^2}{\partial P_2^2} + V(r_1, -r_2) \right] \Psi(t) \quad (14)$$

Substituting equ (9) & (13) into equ (14) yields;

$$i\hbar \frac{\partial \Psi(R, P, t)}{\partial t} = \left[\frac{-\hbar^2}{2m_2} \left[\frac{\partial^2}{\partial P^2} + \left(\frac{m_2}{m_1 + m_2} \right)^2 \frac{\partial^2}{\partial R^2} + \frac{2m_2}{m_1 + m_2} \frac{\partial}{\partial R} \frac{\partial}{\partial P} \right] - \frac{\hbar^2}{2m_1} \left[\frac{\partial^2}{\partial P^2} + \left(\frac{m_1}{m_1 + m_2} \right)^2 \frac{\partial^2}{\partial R^2} - \frac{2m_1}{m_1 + m_2} \frac{\partial}{\partial R} \frac{\partial}{\partial P} \right] + V(r_1, -r_2) \right] \Psi(t) \quad (15)$$

$$i\hbar \frac{\partial \psi}{\partial t} = \left\{ \frac{-\hbar^2}{2} \left(\frac{1}{m_1} + \frac{1}{m_2} \right) \frac{\partial^2}{\partial p^2} - \frac{\hbar^2}{2} \left[\frac{1}{m_2} \left(\frac{m_2}{m_1+m_2} \right)^2 + \frac{1}{m_1} \left(\frac{m_1}{m_1+m_2} \right)^2 \right] \frac{\partial^2}{\partial R^2} \right\} \psi$$

$$\left[\frac{-\hbar^2}{2m_2} \left[\frac{2m_2}{m_1+m_2} \right] + \frac{\hbar^2}{2m_1} \left[\frac{2m_1}{m_1+m_2} \right] \right] \frac{\partial^2}{\partial R^2} \psi$$

$$i\hbar \frac{\partial \psi}{\partial t} = \left\{ \frac{-\hbar^2}{2} \left(\frac{m_1+m_2}{m_1 m_2} \right) \frac{\partial^2}{\partial p^2} - \frac{\hbar^2}{2} \left(\frac{m_2}{m_1+m_2} + \frac{m_1}{m_1+m_2} \right) \frac{\partial^2}{\partial R^2} + V(r_1, r_2) \right\} \psi(t) \quad (16)$$

$$\text{Let } (M = m_1 + m_2) \text{ and } \left(\frac{1}{M} = \frac{m_1}{m_1+m_2} + \frac{m_2}{m_1+m_2} \right) \quad (17)$$

$$i\hbar \frac{\partial \psi}{\partial t} = \frac{-\hbar^2}{2} \left(\frac{m_1+m_2}{m_1 m_2} \right) \frac{\partial^2}{\partial p^2} - \frac{\hbar^2}{2} \left(\frac{m_1+m_2}{m_1+m_2} \right) \frac{\partial^2}{\partial R^2} + V(r_1, r_2) \quad (18)$$

$$i\hbar \frac{\partial \psi}{\partial t} = \frac{-\hbar^2}{2} \left(\frac{1}{M} \right) \frac{\partial^2}{\partial p^2} - \frac{\hbar^2}{2} \left(\frac{1}{M} \right) \frac{\partial^2}{\partial R^2} + V(r_1, r_2) \quad (19)$$

$$\therefore i\hbar \frac{\partial \psi}{\partial t} (p, R, t) = \left\{ \frac{-\hbar^2}{2M} \frac{\partial^2}{\partial p^2} - \frac{\hbar^2}{2M} \frac{\partial^2}{\partial R^2} + V(r_1, r_2) \right\} \psi \quad (20)$$

$\therefore$ equ (21) is the time dependent schrodinger equation with reduced mass; $\left(\frac{1}{M} = \frac{m_1}{m_1+m_2} + \frac{m_2}{m_1+m_2} \right)$

2. state the quantum conditions & Properties of the Commutator;

Solu.

Non-Commutativity $[A, B] = AB - BA \neq 0$

Anti-Symmetry $[A, B] = -[B, A]$

Canonical Commutation Relation $[x, p] = i\hbar$ and $[p, x] = -i\hbar$

$[A, BC] = [A, B]C + B[A, C]$

$[AB, C] = A[B, C] + [A, C]B \rightarrow \text{LINEARITY.}$

Properties.

- An operator commutes with a scalar $[A, a] = 0$
- An operator commutes with itself $[A, A] = 0$.
- An operator commutes with any power of itself; $[A, A^n] = 0$.

(Q.11) $[S_x, S_y] = i\hbar S_z, [S_y, S_z] = i\hbar S_x, [S_z, S_x] = i\hbar S_y$

Solu.

→ recall; $\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$

$\sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$

Also $S = \frac{\hbar}{2} \sigma$

$$\Rightarrow S_x = \frac{\hbar}{2} \sigma_x, S_y = \frac{\hbar}{2} \sigma_y, S_z = \frac{\hbar}{2} \sigma_z.$$

$$[S_x, S_y] = S_x S_y - S_y S_x.$$

$$\left(\frac{\hbar}{2} \sigma_x \cdot \frac{\hbar}{2} \sigma_y \right) - \left(\frac{\hbar}{2} \sigma_y \cdot \frac{\hbar}{2} \sigma_x \right)$$

$$= \frac{\hbar^2}{4} (\sigma_x \cdot \sigma_y) - \frac{\hbar^2}{4} (\sigma_y \cdot \sigma_x)$$

$$\frac{\hbar^2}{4} [\sigma_x \sigma_y - \sigma_y \sigma_x]$$

$$= \frac{\hbar^2}{4} \left[\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} - \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \right]$$

$$= \frac{\hbar^2}{4} \left[\begin{pmatrix} 0(0) + 1(i) & 0(-i) + 1(0) \\ 1(0) + 0(i) & 1(-i) + 0(0) \end{pmatrix} - \begin{pmatrix} 0(0) - i(1) & 0(1) - i(0) \\ i(0) + 0(i) & i(1) + 0(0) \end{pmatrix} \right]$$

$$= \frac{\hbar^2}{4} \left[\begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix} - \begin{pmatrix} -i & 0 \\ 0 & i \end{pmatrix} \right]$$

$$= \frac{\hbar^2}{4} \begin{bmatrix} i - (-i) & 0 - 0 \\ 0 - 0 & -i - i \end{bmatrix} = \frac{\hbar^2}{4} \begin{bmatrix} 2i & 0 \\ 0 & -2i \end{bmatrix}$$

$$= \frac{\hbar^2}{4} (2i) \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$[S_x, S_y] = \frac{\hbar^2}{2} i \sigma_z.$$

recall that; $S_z = \frac{\hbar}{2} \sigma_z$

$$\sigma_z = \frac{2S_z}{\hbar}$$

$$2 \quad [s_x, s_y] = \frac{\hbar^2}{2} i \left(\frac{2s_z}{\hbar} \right)$$

$$\therefore [S_x, S_y] = i\hbar S_z \quad \text{u.}$$

$$\Rightarrow [S_y, S_z] = i\hbar S_x.$$

$$a [s_y, s_z]_a = \frac{\hbar^2}{4} [\sigma_y \sigma_z - \sigma_z \sigma_y]$$

$$= \frac{\hbar^2}{4} \left[\begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} - \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix} \right]$$

$$\frac{t_n^2}{4} \begin{pmatrix} (v(0) - i(0)) & (v(0) - i(-i)) \\ i(1) + v(0) & (v(0) + v(-i)) \end{pmatrix} - \begin{pmatrix} (i(0) + v(i)) & (i(-i) + v(0)) \\ v(0) - i(i) & v(0) - i(-i) \end{pmatrix}$$

$$\frac{\hbar^2}{4} \left[\begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix} - \begin{pmatrix} 0 & -i \\ -i & 0 \end{pmatrix} \right]$$

$$\frac{t_1^2}{4} \begin{bmatrix} (0 \neq 0) & i - (-i) \\ i - (-i) & 0 - 0 \end{bmatrix}$$

$$= \frac{t^2}{4} \begin{bmatrix} 0 & 2i \\ 2i & 0 \end{bmatrix}$$

$$\frac{t^2}{4} (2i) \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$= \frac{t^2}{2} : \sigma_x$$

recall that $s_x = \frac{t}{2} \sigma_x$

$$\sigma_x = \cancel{2S_x/t}$$

Emeka Nobis aufb

$$[S_y, S_z] = \frac{\hbar^2}{2} i \left(\frac{2S_x}{\hbar} \right)$$

$$[S_y, S_z] = i\hbar S_x$$

$\Rightarrow [S_z, S_x] = i\hbar S_y$

$$[S_z, S_x] = \frac{\hbar^2}{4} [\sigma_z \sigma_x - \sigma_x \sigma_z]$$

$$= \frac{\hbar^2}{4} \left[\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} - \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \right]$$

$$= \frac{\hbar^2}{4} \left[\begin{pmatrix} 1(0) + 0(1) & 1(1) + 0(0) \\ 0(0) - 1(1) & 0(1) - 1(0) \end{pmatrix} - \begin{pmatrix} 0(1) + 1(0) & 0(0) + 1(-1) \\ 1(1) + 0(0) & 1(0) + 0(-1) \end{pmatrix} \right]$$

$$= \frac{\hbar^2}{4} \left[\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} - \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \right]$$

$$= \frac{\hbar^2}{4} \begin{bmatrix} (0-0) & 1-(-1) \\ (-1-1) & (0-0) \end{bmatrix}$$

$$= \frac{\hbar^2}{4} \begin{bmatrix} 0 & 2 \\ -2 & 0 \end{bmatrix}$$

$$= \frac{\hbar^2}{4} (2i) \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

$$= \frac{\hbar^2}{2} i \sigma_y$$

recall that $S_y = \frac{\hbar}{2} \sigma_y$

$$\sigma_y = \frac{2S_y}{\hbar}$$

$$[S_z, S_x] = \frac{\hbar^2}{2} i \left(\frac{2S_y}{\hbar} \right)$$

$$[S_z, S_x] = i\hbar S_y$$

(2b) using Pauli matrices show that;

$$(\sigma \cdot A)(\sigma \cdot B) = (A \cdot B)I + i\sigma \cdot (A \times B)$$

$$\sigma = \sigma_x + \sigma_y + \sigma_z$$

$$A = A_x + A_y + A_z$$

$$B = B_x + B_y + B_z$$

$$(\sigma \cdot A) \rightarrow (\sigma_x A_x + \sigma_y A_y + \sigma_z A_z)$$

$$(\sigma \cdot B) \rightarrow (\sigma_x B_x + \sigma_y B_y + \sigma_z B_z)$$

$$(\sigma \cdot A)(\sigma \cdot B) = (\sigma_x A_x + \sigma_y A_y + \sigma_z A_z)(\sigma_x B_x + \sigma_y B_y + \sigma_z B_z)$$

$$= \sigma_x^2 A_x B_x + \sigma_x \sigma_y A_x B_y + \sigma_x \sigma_z A_x B_z + \sigma_y \sigma_x A_y B_x + \sigma_y^2 A_y B_y + \sigma_y \sigma_z A_y B_z + \sigma_z \sigma_x A_z B_x + \sigma_z \sigma_y A_z B_y + \sigma_z^2 A_z B_z$$

Note;

$$\sigma_x^2 = 1 ; \sigma_y^2 = 1 ; \sigma_z^2 = 1$$

$$\sigma_x \sigma_y = i\sigma_z$$

$$\sigma_y \sigma_z = i\sigma_x$$

$$\sigma_z \sigma_x = i\sigma_y$$

$$\sigma_y \sigma_x = -i\sigma_z$$

$$\sigma_z \sigma_y = -i\sigma_x$$

$$\sigma_x \sigma_z = -i\sigma_y$$

Take note;
To be substituted

$$(\sigma \cdot A)(\sigma \cdot B) = A_x B_x + i\sigma_z A_x B_y - i\sigma_y A_x B_z - i\sigma_z A_y B_x + A_y B_y + i\sigma_x A_y B_z + i\sigma_y A_z B_x - i\sigma_x A_z B_y + A_z B_z$$

$$= A_x B_x + A_y B_y + A_z B_z + i(\sigma_x A_y B_z + \sigma_y A_z B_x + \sigma_z A_x B_y) - i(\sigma_y A_x B_z + \sigma_z A_y B_x + \sigma_x A_z B_y).$$

$$(\sigma \cdot A)(\sigma \cdot B) = A \cdot B + i\sigma \cdot (A \times B).$$

where; $A \cdot B \rightarrow A_x B_x + A_y B_y + A_z B_z$

$$i\sigma \cdot (A \times B) \rightarrow \begin{vmatrix} i\sigma_x & i\sigma_y & i\sigma_z \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$$

4

A(i) what do you understand by identical particles?

⇒ Particles of a system are identical if no observer can detect any permutation of these particles.

A(ii) Any arbitrary permutation σ is said to be symmetrical if $\sigma|u\rangle = |u\rangle$. The same vector is said to be anti-symmetrical if $\sigma|u\rangle = (\text{sgn}\sigma)|u\rangle$
 Bosons → Completely symmetrical wave function
 Fermions → Anti-symmetrical wave functions.

4Aiii Symmetrization Postulate.

The Symmetrization postulate for fermions states that the wave function of a system of n identical fermions is completely ~~antisymmetric~~ symmetric. Thus, it is a combination of vectors of the form $|\alpha_j, \dots, j_n\rangle$. These normalized vectors can be written as;

$$|\alpha_j, \dots, j_n\rangle = \frac{1}{\sqrt{n!}} \begin{vmatrix} |\phi_1^{j_1}\rangle & |\phi_1^{j_2}\rangle & \dots & |\phi_1^{j_n}\rangle \\ |\phi_2^{j_1}\rangle & |\phi_2^{j_2}\rangle & \dots & |\phi_2^{j_n}\rangle \\ \vdots & \vdots & \ddots & \vdots \\ |\phi_n^{j_1}\rangle & \dots & \dots & |\phi_n^{j_n}\rangle \end{vmatrix}$$

Hence, if two particles are in the same quantum state, two columns are the same, forcing the determinant to vanish. This restriction on their wavefunction leads to Pauli's exclusion Principle.

4Aiv The state's determinant for two fermions is given by;

$$|\psi_{(1,2)}\rangle = \frac{1}{\sqrt{2!}} \begin{vmatrix} |\phi_1^{j_1}\rangle & |\phi_1^{j_2}\rangle \\ |\phi_2^{j_1}\rangle & |\phi_2^{j_2}\rangle \end{vmatrix}$$

$$= \frac{1}{\sqrt{2!}} [(\phi_1^{j_1})(\phi_2^{j_2}) - (\phi_1^{j_2})(\phi_2^{j_1})]$$

$$|u_{(2,1)}\rangle = \frac{1}{\sqrt{2!}} \begin{vmatrix} |\Phi_2^{j_1}\rangle & |\Phi_2^{j_2}\rangle \\ |\Phi_1^{j_1}\rangle & |\Phi_1^{j_2}\rangle \end{vmatrix}$$

$$= \frac{1}{\sqrt{2!}} [(\Phi_2^{j_1}) (\Phi_1^{j_2}) - (\Phi_2^{j_2}) (\Phi_1^{j_1})]$$

If we multiply through by $-i$;

$$\frac{1}{\sqrt{2!}} [(\Phi_2^{j_2}) (\Phi_1^{j_1}) - (\Phi_2^{j_1}) (\Phi_1^{j_2})]$$

$$|u_{(2,2)}\rangle = -|u_{(2,1)}\rangle$$

(5)

Q If $(q, p) : a' = a'(q, p)$ find q and p .

$$a = (2m\omega\hbar)^{-1/2} (m\omega q + i p)$$

$$a^\dagger = (2m\omega\hbar)^{-1/2} (m\omega q - i p)$$

$$\text{let } A = (2m\omega\hbar)^{-1/2}$$

$$a = A(m\omega q + i p) = Am\omega q + iAP$$

$$a^\dagger = A(m\omega q - i p) = Am\omega q - iAP$$

$$a + a^\dagger = 2Am\omega q$$

$$q = \frac{a + a^\dagger}{2Am\omega}$$

$$\text{recall; } A = (2m\omega\hbar)^{-1/2}$$

$$\therefore q = \frac{(2m\omega\hbar)^{1/2} (a + a^\dagger)}{2Am\omega}$$

$$a - a^\dagger = 2iAP$$

$$P = \frac{a - a^\dagger}{2iA}$$

multiplying through by i ;

$$\left(P = \frac{a - a^\dagger}{2iA} \right) \times \frac{i}{i}$$

$$P = \frac{-i(a - a^\dagger)}{2A}$$

$$P = \frac{i(a^\dagger - a)}{2A}$$

recall that: $A = (2m\omega\hbar)^{-1/2}$

$$P = \frac{i(2m\omega\hbar)^{1/2}}{2} (a^\dagger - a)$$

5B. $\Rightarrow [a, a] = aa - aa$
 If $a = (2m\omega\hbar)^{-1/2} (m\omega q + i p)$
 Let $A = (2m\omega\hbar)^{-1/2}$
 $= A^2 [(m\omega q + i p)(m\omega q + i p) - (m\omega q + i p)(m\omega q + i p)]$
 $\therefore [a, a] = 0$

$\Rightarrow [a^\dagger, a^\dagger] = 0$
 $a^\dagger = A(m\omega q - i p)$
 $(a^\dagger a^\dagger - a^\dagger a^\dagger) = A^2 [(m\omega q - i p)(m\omega q - i p) - (m\omega q - i p)(m\omega q - i p)]$
 $= A^2 [m^2\omega^2 q^2 - im\omega p q - im\omega p q + p^2 - m^2\omega^2 q^2 - im\omega p q - im\omega p q + p^2]$
 $\therefore [a^\dagger, a^\dagger] = 0$

$\Rightarrow [a, a^\dagger] = (a a^\dagger - a^\dagger a)$
 $a a^\dagger = (2m\omega\hbar)^{-1/2} (m\omega q + i p) \cdot (2m\omega\hbar)^{-1/2} (m\omega q - i p)$
 $= (2m\omega\hbar)^{-1/2}^2 (m\omega q + i p)(m\omega q - i p)$
 $= \frac{1}{2m\omega\hbar} [m^2\omega^2 q^2 - im\omega q p + im\omega p q + p^2]$

$$= \frac{1}{2m\omega\hbar} \left[p^2 + m^2\omega^2 q^2 + im\omega(pq - qp) \right]$$

$$= \frac{1}{\omega\hbar} \left[\frac{p^2}{2m} + \frac{m^2\omega^2 q^2}{2m} + \frac{im\omega}{2m} (pq - qp) \right]$$

$$= \frac{1}{\omega\hbar} \left[\frac{p^2}{2m} + \frac{m\omega^2 q^2}{2} + \frac{i\omega}{2} (pq - qp) \right]$$

recall ; $H = \frac{p^2}{2m} + \frac{1}{2} m\omega^2 q^2$

and $[p, q] = -i\hbar = (pq - qp)$

$$= \frac{1}{\omega\hbar} \left[\frac{H + \frac{i\omega}{2} (-i\hbar)}{2} \right] = \frac{1}{\omega\hbar} \left[\frac{H + \frac{\omega\hbar}{2}}{2} \right]$$

Also;

$$a^\dagger a = (2m\omega\hbar)^{1/2} (m\omega q - ip) \cdot (2m\omega\hbar)^{1/2} (m\omega q + ip)$$

$$= \frac{1}{2m\omega\hbar} \left[m^2\omega^2 q^2 + im\omega qp - im\omega pq + p^2 \right]$$

$$= \frac{1}{2m\omega\hbar} \left[p^2 + m^2\omega^2 q^2 + im\omega (qp - pq) \right]$$

$$= \frac{1}{\omega\hbar} \left[\frac{p^2}{2m} + \frac{m^2\omega^2 q^2}{2m} + \frac{im\omega}{2m} (qp - pq) \right]$$

But ; $H = \frac{p^2}{2m} + \frac{1}{2} m\omega^2 q^2$

and $[q, p] = i\hbar$

$$= \frac{1}{\omega\hbar} \left[\frac{H + \frac{i\omega}{2} (i\hbar)}{2} \right]$$

$$= \frac{1}{\omega \hbar} \left[H - \frac{\omega \hbar}{2} \right]$$

$$\therefore [a, a^\dagger] = a a^\dagger - a^\dagger a$$

$$= \frac{1}{\omega \hbar} \left[H + \frac{\omega \hbar}{2} \right] - \frac{1}{\omega \hbar} \left[H - \frac{\omega \hbar}{2} \right]$$

$$= \frac{H}{\omega \hbar} + \frac{1}{2} - \frac{H}{\omega \hbar} + \frac{1}{2} = 1$$

show that; $H = \left(N + \frac{1}{2}\right) \omega \hbar$ where $N = a^\dagger a$.

→ If $N = a^\dagger a$

recall $a^\dagger a = \frac{1}{\omega \hbar} \left[H - \frac{\omega \hbar}{2} \right]$

$$N = \frac{H}{\omega \hbar} - \frac{1}{2}$$

$$N = \frac{H - \frac{\omega \hbar}{2}}{\omega \hbar}$$

$$N = \frac{2H - \omega \hbar}{2\omega \hbar}$$

$$2N\omega \hbar = 2H - \omega \hbar$$

$$2H = 2N\omega \hbar + \omega \hbar$$

$$2H = \omega \hbar (2N + 1)$$

$$H = \frac{\omega \hbar (2N + 1)}{2}$$

$$H = \omega \hbar \left(N + \frac{1}{2} \right)$$

5D

show that $[a, N] = a$, $[a^\dagger, N] = -a^\dagger$.

where $N = a^\dagger a$

$$\Rightarrow [a, N] = [a, a^\dagger a] = [a, a^\dagger]a + [a, a]a^\dagger$$

where;

$$[a, a^\dagger] = 1$$

$$[a, a] = 0$$

$$\Rightarrow [a, N] = (1)a + 0 = a. \text{ Proved.}$$

$$\Rightarrow [a^\dagger, N] = -a^\dagger = [a^\dagger, a^\dagger a]$$

$$= [a^\dagger, a^\dagger]a + [a^\dagger, a]a^\dagger$$

where; $a^\dagger a^\dagger = 0$

$$a^\dagger a = -1$$

$$[a^\dagger, N] = 0 + (-1)a^\dagger = -a^\dagger. \text{ Proved.}$$

(SE)

(i) $H(a|E')>$

(ii) $H(a^+|E')>$

soln.

(i) $H(a|E')>$

recall: $H = (N + \frac{1}{2}) \omega \hbar$

$$H(a|E')> = (N + \frac{1}{2}) \omega \hbar (a|E')>$$

$$[H, a] = (Na + \frac{1}{2}a)|E')> \omega \hbar$$

$$[H, a] + a[H, N] = (Na|E')> + \frac{1}{2}a|E')> \omega \hbar$$

(ii) $H(a^+|E')>$

since $H = (N + \frac{1}{2}) \omega \hbar$

$$[N, a^+] = \frac{H}{\omega \hbar} - \frac{1}{2} \quad \text{--- (1)}$$

since $[a, N] = aN - Na = a$

$$[a^+, N] = a^+N - Na^+ = -a^+ \quad \text{(2)}$$

$$Na^+ = a^+N + a^+ = a^+ \left(\frac{H}{\omega \hbar} - \frac{1}{2} \right) + a^+ \quad \text{(3)}$$

$$H(a^+|E')> = (N + \frac{1}{2}) \omega \hbar (a^+|E')>$$

$$= (Na^+|E')> + \frac{1}{2}a^+|E')> \omega \hbar$$

substituting eqn (2) into (3);

$$\left\{ \left[a^\dagger \left(\frac{H}{\hbar\omega} - \frac{1}{2} \right) + a^\dagger \right] |E'\rangle + \frac{1}{2} a^\dagger |E'\rangle \right\} \hbar\omega$$

$$\left\{ \frac{a^\dagger H}{\hbar\omega} |E'\rangle - \frac{1}{2} a^\dagger |E'\rangle + a^\dagger |E'\rangle + \frac{1}{2} a^\dagger |E'\rangle \right\} \hbar\omega$$

$$\left\{ \frac{a^\dagger H}{\hbar\omega} |E'\rangle + a^\dagger |E'\rangle \right\} \hbar\omega$$

$$\left\{ \frac{a^\dagger E'}{\hbar\omega} |E'\rangle + a^\dagger |E'\rangle \right\} \hbar\omega$$

$$= a^\dagger E' |E'\rangle + a^\dagger \hbar\omega |E'\rangle$$

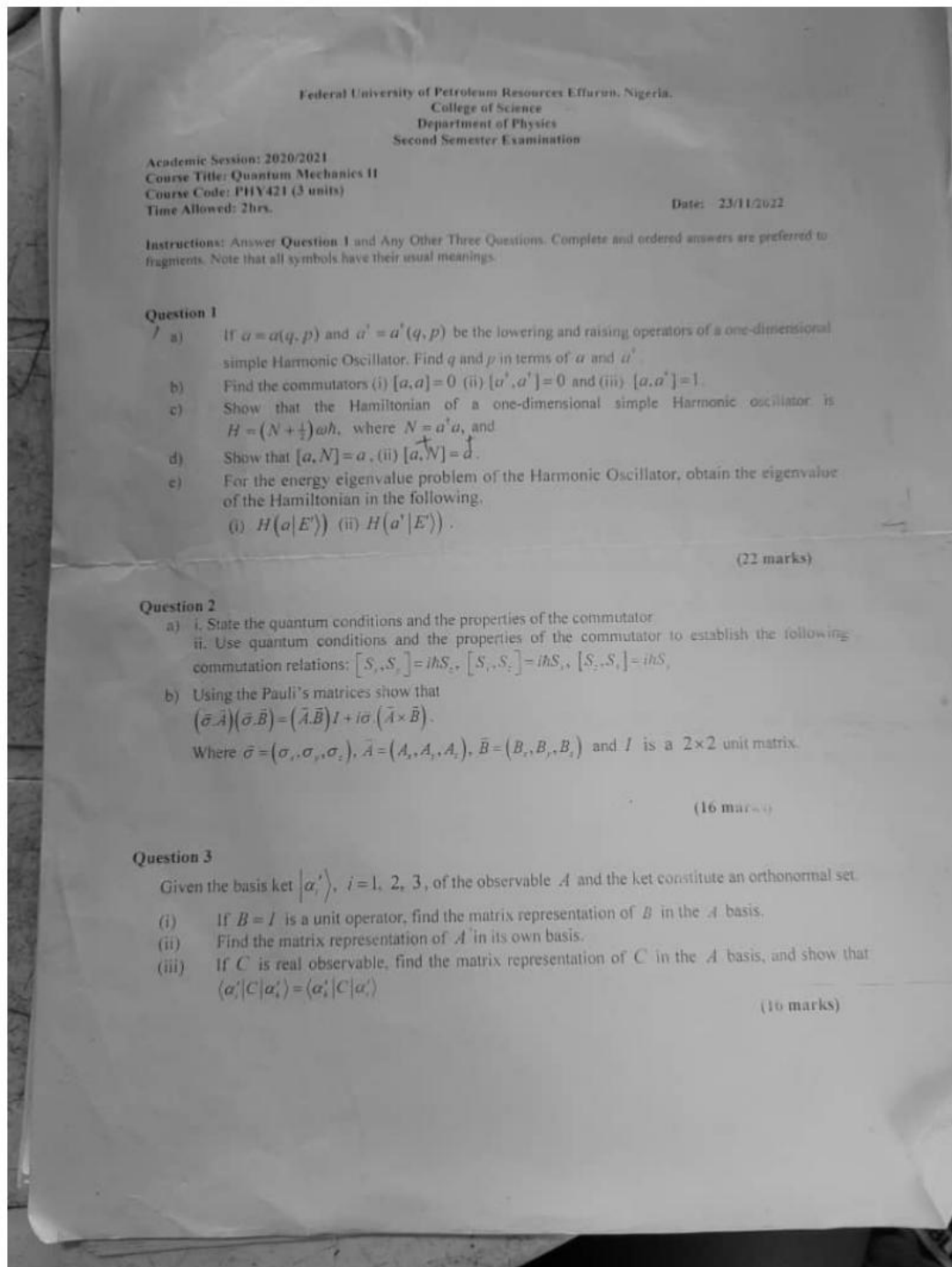
$$= (E' + \hbar\omega) a^\dagger |E'\rangle$$

where $(E' + \hbar\omega)$ is the eigen value acting on ket $|E'\rangle$.

Paper 4 of 4 — Phy 421 PQ QUANTUM MECHANICS II

Level 400 • Second Semester • PastQuestion

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Question 4

- (i) Prove that if $i = k$, all the diagonal elements of C (in Question 3 (iii)) are real, and each of the other elements is equal to the complex conjugate of its mirror image in the diagonal, the matrix is then called the adjoint matrix.
- (ii) If $D = BC$, establish that $\langle \alpha' | D | \alpha' \rangle = \sum_i \langle \alpha' | C | \alpha'_i \rangle \langle \alpha'_i | B | \alpha' \rangle$ which is just the law of matrix multiplication. The multiplication of matrices is non-commutative just like the multiplication of linear operators. Where matrix B and C are the same as defined in Question 3 above

(16 marks)

Question 5

- i. What do you understand by identical particles?
- ii. Write down the symmetric and antisymmetric properties for any arbitrary permutation σ of identical particles. Give one example each of particles with completely symmetrical wave functions and completely antisymmetrical wave functions.

(16 marks)

Question 6

- i. Use the symmetrization postulate for fermions to derive the Pauli exclusion principle.
- ii. Show explicitly that Slater's determinant for two particles (fermions) is antisymmetric.

(16 marks)